

SHETH L.U.J. AND SIR M.V. COLLEGE
PRACTICAL NO.:-5
SUB:- CYBER SECURITY

AIM:-Implement the Diffie-Hellman key exchange algorithm to securely exchange keys between two entities over an insecure network.

INPUT:-

```
print("DIFFIE-HELLMAN KEY EXCHANGE")
print("-----")

p = int(input("Enter prime number (p): "))
g = int(input("Enter primitive root (g): "))

a = int(input("Enter Vibhu's private key: "))
b = int(input("Enter Unna's private key: "))

A = pow(g, a, p)
B = pow(g, b, p)

key_Vibhu = pow(B, a, p)
key_Unna = pow(A, b, p)

print("\n--- RESULTS ---")

print("Vibhu's Public Key:", A)
print("Unna's Public Key:", B)

print("Vibhu's Shared Secret Key:", key_Vibhu)
print("Unna's Shared Secret Key:", key_Unna)

if key_Vibhu == key_Unna:
    print("\nKey Exchange Successful!")
    print("Both Vibhu and Unna have the same secret key.")
else:
    print("\nKey Exchange Failed!")
```

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OUTPUT:-

```
D:\T079_VIBHUTI > CYBER SECURITY > PRACS_VIBHUTI.py > ...
12 b = pow(g, b, p)
13
14 key_Vibhu = pow(B, a, p)
15 key_Unna = pow(A, b, p)
16
17 print("\n--- RESULTS ---")
18
19 print("Vibhu's Public Key:", A)
20 print("Unna's Public Key:", B)
21
22 print("Vibhu's Shared Secret Key:", key_Vibhu)
23 print("Unna's Shared Secret Key:", key_Unna)
24
25
26
27 if key_Vibhu == key_Unna:
28     print("\nKey Exchange Successful!")
29     print("Both Vibhu and Unna have the same secret key.")
30 else:
31     print("Key Exchange Failed!")
32
```

```
Enter prime number (p):
PS D:\T079_VIBHUTI\CYBER SECURITY> d:; cd 'd:\T079_VIBHUTI\CYBER SECURITY'; & 'C:\Users\mvlu\AppData\Local\Programs\Python\Python314\python.exe' 'c:\Users\mvlu\
1\vscode\extensions\ms-python.python-2023.4.1\pythonFiles\lib\python\debugpy\adapter\..\..\debugpy\launcher' '47458' '-' 'd:\T079_VIBHUTI\CYBER SECURITY\PRACS_
VIBHUTI.py'
DIFFIE-HELLMAN KEY EXCHANGE
-----
Enter prime number (p): 13
Enter primitive root (g): 5
Enter Vibhu's private key: 1
Enter Unna's private key: 24

--- RESULTS ---
Vibhu's Public Key: 5
Unna's Public Key: 1
Vibhu's Shared Secret Key: 1
Unna's Shared Secret Key: 1

Key Exchange Successful!
Both Vibhu and Unna have the same secret key.
PS D:\T079_VIBHUTI\CYBER SECURITY>
```

GUI

INPUT:-

```
import tkinter as tk
from tkinter import messagebox

def calculate_keys():
    try:
        p = int(p_entry.get())
        g = int(g_entry.get())
        a = int(vibhu_entry.get())
        b = int(unna_entry.get())

        A = pow(g, a, p)
        B = pow(g, b, p)

        key_Vibhu = pow(B, a, p)
        key_Unna = pow(A, b, p)

        vibhu_public_label.config(
            text="Vibhu's Public Key: " + str(A)
        )
    except ValueError:
        messagebox.showerror("Error", "Please enter valid integers.")
```

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```
)

unna_public_label.config(
    text="Unna's Public Key: " + str(B)
)

vibhu_shared_label.config(
    text="Vibhu's Shared Secret Key: " + str(key_Vibhu)
)

unna_shared_label.config(
    text="Unna's Shared Secret Key: " + str(key_Unna)
)

if key_Vibhu == key_Unna:
    result_label.config(
        text="Key Exchange Successful!\n"
        "Both Vibhu and Unna have the same secret key.",
        fg="green"
    )
else:
    result_label.config(
        text="Key Exchange Failed!",
        fg="red"
    )

except ValueError:
    messagebox.showerror(
        "Error",
        "Please enter valid numbers."
    )

def clear_all():
    p_entry.delete(0, tk.END)
    g_entry.delete(0, tk.END)
    vibhu_entry.delete(0, tk.END)
    unna_entry.delete(0, tk.END)

    vibhu_public_label.config(text="")
```

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```
unna_public_label.config(text="")
vibhu_shared_label.config(text="")
unna_shared_label.config(text="")
result_label.config(text="")

root = tk.Tk()
root.title("Diffie-Hellman Key Exchange")
root.geometry("650x650")
root.resizable(False, False)

title = tk.Label(
    root,
    text="DIFFIE-HELLMAN KEY EXCHANGE",
    font=("Arial", 20, "bold")
)
title.pack(pady=20)

tk.Label(
    root,
    text="Enter Prime Number (p):",
    font=("Arial", 12, "bold")
).pack()

p_entry = tk.Entry(
    root,
    width=30,
    font=("Arial", 12)
)
p_entry.pack(pady=5)

tk.Label(
    root,
    text="Enter Primitive Root (g):",
    font=("Arial", 12, "bold")
).pack()

g_entry = tk.Entry(
    root,
    width=30,
```

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```
        font=("Arial", 12)
    )
    g_entry.pack(pady=5)

    tk.Label(
        root,
        text="Enter Vibhu's Private Key:",
        font=("Arial", 12, "bold")
    ).pack()

    vibhu_entry = tk.Entry(
        root,
        width=30,
        font=("Arial", 12)
    )
    vibhu_entry.pack(pady=5)

    tk.Label(
        root,
        text="Enter Unna's Private Key:",
        font=("Arial", 12, "bold")
    ).pack()

    unna_entry = tk.Entry(
        root,
        width=30,
        font=("Arial", 12)
    )
    unna_entry.pack(pady=5)

    tk.Button(
        root,
        text="Generate Shared Key",
        command=calculate_keys,
        font=("Arial", 11, "bold"),
        padx=20,
        pady=7
    ).pack(pady=15)
```

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```
vibhu_public_label = tk.Label(  
    root,  
    text="",  
    font=("Arial", 11, "bold")  
)  
vibhu_public_label.pack(pady=5)  
  
unna_public_label = tk.Label(  
    root,  
    text="",  
    font=("Arial", 11, "bold")  
)  
unna_public_label.pack(pady=5)  
  
vibhu_shared_label = tk.Label(  
    root,  
    text="",  
    font=("Arial", 11, "bold")  
)  
vibhu_shared_label.pack(pady=5)  
  
unna_shared_label = tk.Label(  
    root,  
    text="",  
    font=("Arial", 11, "bold")  
)  
unna_shared_label.pack(pady=5)  
  
result_label = tk.Label(  
    root,  
    text="",  
    font=("Arial", 11, "bold"),  
    justify="center"  
)  
result_label.pack(pady=20)  
  
tk.Button(  
    root,  
    text="Clear",
```

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```
command=clear_all,  
font=("Arial", 10),  
padx=25  
) .pack()  
  
root.mainloop()
```

OUTPUT:-

